

Introduction to Machine Learning: From Theory to Application

About Me

- ML PhD Student at Georgia Tech
- What I have done:



Understanding Fitbit
Data

Stanford
Continuing
Studies

Machine Learning with
Python Instructor

LinkedIn Learning

LinkedIn Learning
Instructor

What you should know for this talk

- Linear regression
- Python strings
- Integers and floats
- Conditionals and booleans-if, else, and elif
- For loops
- Lists, tuples
- Dictionaries

What is Machine Learning?



“field of study that gives computers the ability to learn from data without being explicitly programmed.”

- Arthur Samuel

What is Machine Learning?



“‘Learning’ enters when we give these models tunable parameters that can be adapted to observed data; in this way the program can be considered to be ‘learning’ from the data.”

- Jake Vanderplas

What is Machine Learning?



“Once models have been fit to previously seen data, they can be used to predict and understand aspects of newly observed data.”

- Jake Vanderplas

Two Basic Types of Machine Learning

Supervised learning

Modeling the relationship between measured features of data and some target (label/value). When target is a label, it is classification. When target is a continuous value, it is regression.

Unsupervised learning

Modeling the measured features of data without some target (label/value).

Supervised learning

Features Matrix

Number of Columns

Number of Rows

Target Vector

Number of Rows

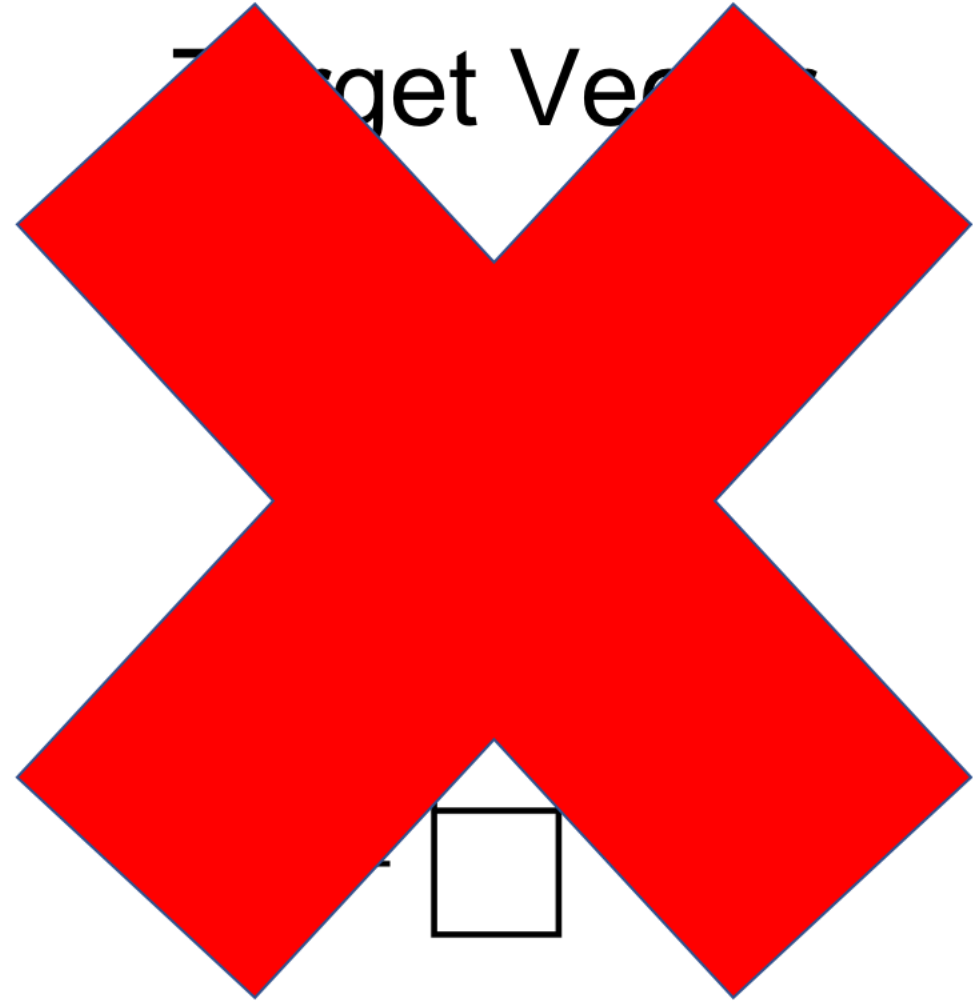
Unsupervised learning

Features Matrix

Number of Columns

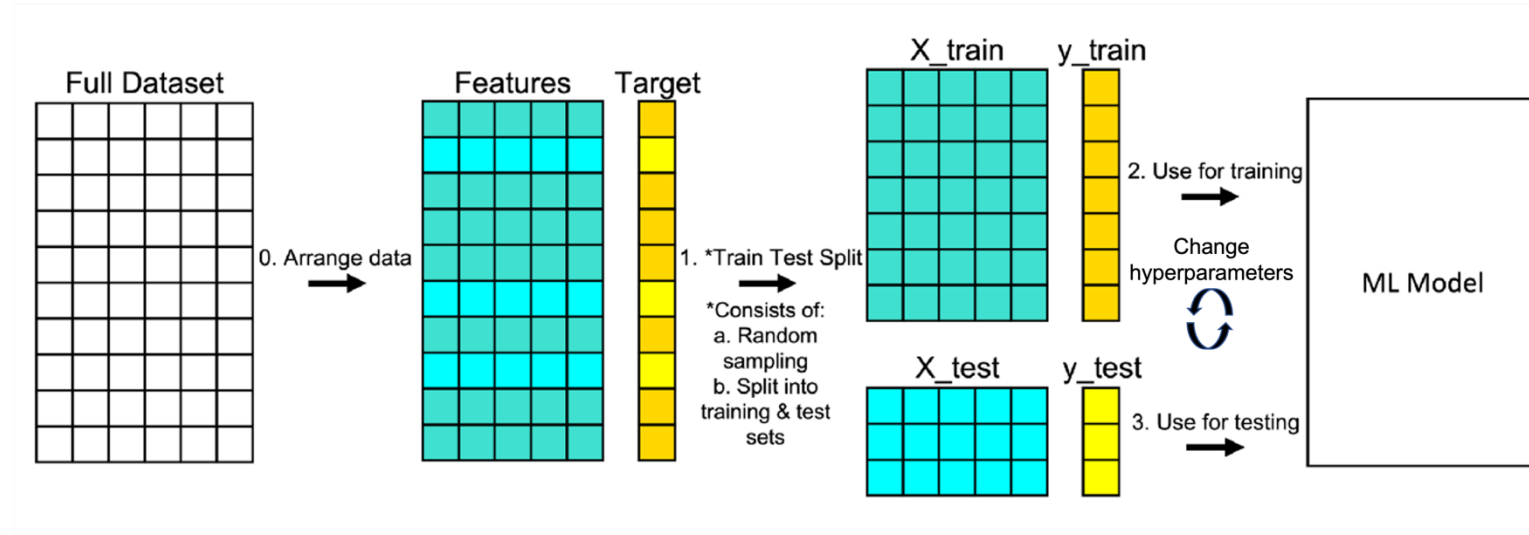
Number of Rows

Target Vector



From Theory to Application

- **A Little Theory + Diagrams gives us the why:** understanding how models learn patterns from data.



From Theory to Application

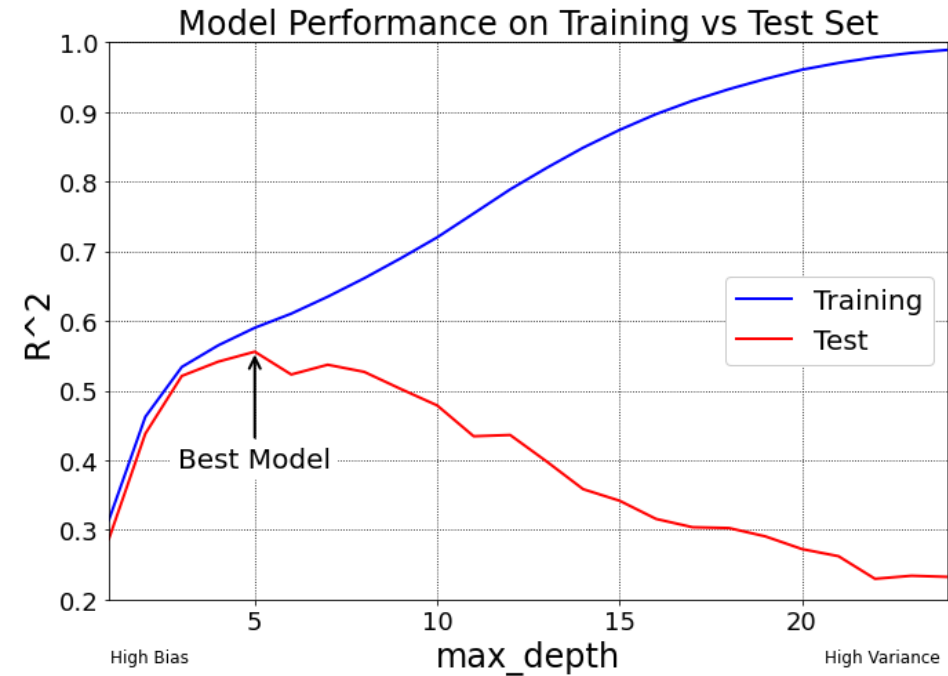
- **Application shows the how:** turning that understanding into working code.

```
max_depth_range = list(range(1, 25))
# List to store the average RMSE for each value of max_depth:
r2_list = []
for depth in max_depth_range:
    reg = DecisionTreeRegressor(max_depth = depth,
                                random_state = 0)
    reg.fit(X_train, y_train)

    score = reg.score(X_test, y_test)
    r2_list.append(score)
```

From Theory to Application

- Images will help with explanations



Why use Scikit-Learn for Machine Learning?

- **This is a not a long talk.**
- Programming machine learning algorithms from scratch is no easy task.
- Scikit-learn provides a large amount of well programmed algorithms.
- Syntax allows you to try multiple algorithms quickly.
- Library works well with other python libraries like NumPy, pandas, and matplotlib.
- Library is popular.

Why about using Cursor/Claude/etc?

- You can use it.
- Understanding how things work will enhance your ability to utilize these tools.